

module2d9

day9

Absolutely 🙌 Since this is for your **final exam**, I'll turn this huge VPN material into **short, high-value exam notes**. I'll separate **what you must memorize**, then give **2 easy + 2 hard examples**.

☒ CISCO VPN – FINAL EXAM NOTES

1. What is a VPN?

VPN = Virtual Private Network

A VPN creates a **secure tunnel** over an untrusted network such as the Internet.

Main purposes

- 🔒 **Confidentiality** → keeps data private
- ☐ **Authentication** → verifies who is communicating
- ❤️ **Integrity** → ensures data was not changed
- ☐ **Secure remote/site connectivity**

Two major VPN types

Type	Meaning	Example
Remote-access VPN	Individual user → company network	Employee working from home
Site-to-site VPN	Network → network	Addis office ↔ Nairobi office

Exam memory:



Remote user = **Remote Access**

Office to office = **Site-to-Site**



2. Important Cisco VPN Technologies

☆ Know these

AnyConnect

- Remote-access VPN client
- Supports **SSL** and **IPsec/IKEv2**
- Used by individual users

FlexVPN

- Modern Cisco site-to-site VPN framework
- Flexible and scalable
- Commonly uses **IKEv2**

DMVPN

- Designed for many branch offices
- Uses **hub-and-spoke** architecture
- Allows dynamic spoke-to-spoke tunnels

Meraki Auto VPN

- Cloud-managed
- Automatically creates VPN tunnels
- Good for many branches

Cisco Secure Firewall (FTD)

- Firewall + VPN + security inspection
- Supports IPsec and SSL VPN



3. Remote Access vs Site-to-Site

Remote Access

Laptop
|
Internet
|
VPN Tunnel
|
Company Network

Copy

Used when **one user/device** connects to the company.

Site-to-Site

LAN A LAN B
PCs – Router — VPN — Router – PCs
Internet

Copy

Used when **two networks** communicate securely.



4. IPsec – VERY IMPORTANT ☆☆☆

IPsec = Internet Protocol Security

It secures IP traffic.

It provides:

- **Authentication**
- **Integrity**
- **Confidentiality**
- **Anti-replay protection**

Important encryption

AES

Especially:



AES-256-GCM

is a modern authenticated-encryption choice.

Avoid old algorithms

Know these:

- × 3DES → deprecated
- × MD5 → weak collision resistance
- × SHA-1 → deprecated for modern security



5. IKEv2

IKE = Internet Key Exchange

IKE negotiates the security parameters used by IPsec.

Think:



IKEv2 = negotiates the VPN security

It establishes things such as:

- Encryption algorithm

- Integrity/authentication algorithm
- Diffie-Hellman group
- Authentication method
- Security associations

Simple memory

IKEv2 → Negotiates security
IPsec → Protects the actual traffic

Copy



6. Perfect Forward Secrecy (PFS)

PFS makes sure that compromising a long-term key does **not automatically reveal old session keys**.

Usually associated with **Diffie-Hellman/ECDHE** key exchange.

Exam definition



PFS protects past sessions even if a long-term private key is later compromised.



7. SSL VPN

SSL VPN allows secure remote access using TLS/SSL technologies.

Advantage

It can work well through restrictive networks/firewalls because HTTPS traffic is commonly allowed.

Clientless SSL VPN

The user can access resources through a **web browser without installing a VPN client**.

Useful for:

- Contractors
- Partners
- BYOD users

Memory



Clientless = Browser-based access



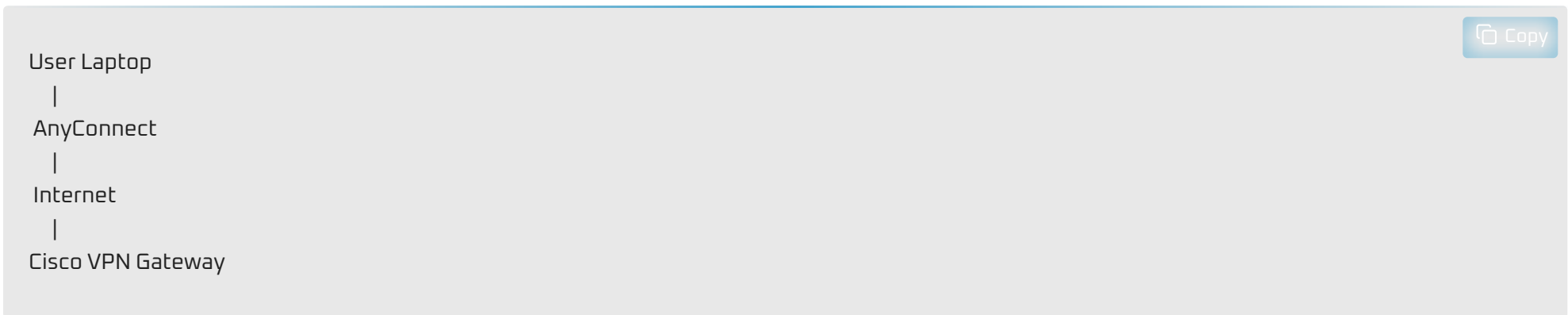
8. AnyConnect ☆

Cisco AnyConnect is a VPN client for remote users.

It can use:

- **SSL/TLS**
- **IPsec/IKEv2**

Simple idea

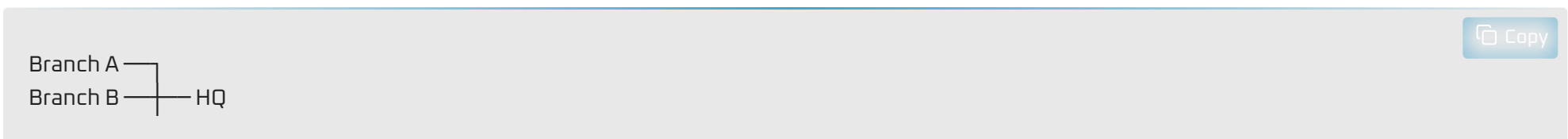


9. DMVPN ☆☆☆

DMVPN = Dynamic Multipoint VPN

Designed for large networks with many branch offices.

Traditional VPN



DMVPN

The hub can help branches establish **dynamic spoke-to-spoke communication**.

Main benefits

- Scalable
- Dynamic tunnels
- Good for many branches
- Reduces need for manually configured tunnels

Exam memory



DMVPN = many branches + dynamic tunnels



10. FlexVPN

FlexVPN is Cisco's flexible VPN framework based heavily around **IKEv2**.

Advantages

- Flexible

- Secure
- Scalable
- Suitable for site-to-site VPN architectures

Exam memory



FlexVPN → flexible IKEv2-based Cisco VPN framework



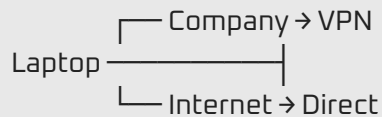
11. Split Tunnel vs Full Tunnel ☆☆☆

This is **very exam-friendly**.

Split tunneling

Only company traffic goes through the VPN.

Other Internet traffic goes directly to the Internet.



Copy

Advantages

- ☐ Less VPN bandwidth
- ☐ Less load on company network
- ☐ Better performance for normal Internet traffic

Disadvantage

× Internet traffic may not receive company security inspection.



Full tunneling

All traffic goes through the VPN.

Laptop
|
VPN
|
Company
|
Internet

Copy

Advantages

- ☐ Centralized security inspection
- ☐ Better control
- ☐ Good for sensitive environments

Disadvantage

- × More bandwidth usage
- × More latency/load

Easy memory



Split = some traffic
Full = everything



12. FTD

FTD = Firepower Threat Defense

Cisco Secure Firewall platform.

Can provide:

- Firewall
- VPN
- IPS/security inspection
- Threat detection

Important concept

FTD can inspect traffic **inside VPN tunnels**.

That is important because encrypted VPN traffic could otherwise hide malicious traffic from security inspection.



13. VPN Encryption vs Authentication

Don't confuse them!

Encryption

Protects **confidentiality**.

Example:

```
HELLO  
↓ encryption  
X7#k92@
```

 Copy

Someone intercepting it cannot easily understand the data.

Authentication

Verifies **identity**.

```
"Are you really the authorized VPN peer?"
```

Integrity

Checks whether the data was modified.

Remember



Encryption = secrecy
Authentication = identity
Integrity = unchanged



14. Quantum-Resistant VPN

This is a newer/future-looking topic.

The idea is to protect VPN cryptography against future **quantum computers**.

Your notes mention:



CRYSTALS-Kyber

and **KEM (Key Encapsulation Mechanism)**.

Exam definition



Quantum-resistant cryptography uses algorithms designed to remain secure against attacks from sufficiently powerful quantum computers.

⚠ **Important:** Don't memorize the presentation's future rollout/performance figures as universal Cisco facts unless your instructor specifically expects those slide values. For an exam, focus on the concept: **post-quantum security + KEM + future quantum resistance**.



15. VPN Protocols / Technologies – Memorize This Table ☆

Technology	Main purpose
IPsec	Secure IP traffic
IKEv2	Negotiates IPsec security
AnyConnect	Remote-access VPN client
SSL VPN	Remote access using TLS/SSL
DMVPN	Scalable dynamic site-to-site VPN
FlexVPN	Flexible IKEv2 VPN framework
Meraki Auto VPN	Automatically managed branch VPN
FTD	Firewall + VPN + security
SASE	Cloud-delivered networking/security architecture





☒ 2 EASY EXAMPLES

Easy Example 1 – Employee Working From Home

Situation

An employee is at home and wants to access the company's internal server.

Solution

Use:



Remote-access VPN

For example:

Employee Laptop



AnyConnect



Internet



VPN Gateway



Company Server

 Copy

Exam answer

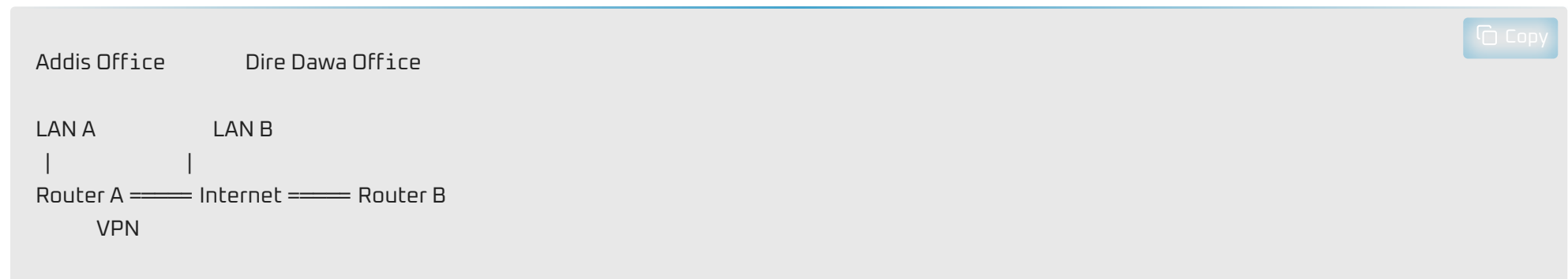
Question: Which VPN type should an individual employee use to securely access a company network remotely?

Answer: ☐ Remote-access VPN



Easy Example 2 – Two Company Offices

Company has:



The company wants secure communication between both offices.

Answer

☐ Site-to-site VPN

Possible Cisco technologies include:

- IPsec
- FlexVPN
- DMVPN

Exam answer



Site-to-site VPN securely connects two or more networks over an untrusted network such as the Internet.



2 HARD EXAM EXAMPLES

Hard Example 1 – Choose Split or Full Tunnel

Question

A bank has remote employees. Security policy requires **all employee Internet traffic to pass through the company's security infrastructure for inspection.**

Should the company use split tunneling or full tunneling?

Answer

☐ **Full tunneling**

Why?

Because:

```
Laptop
↓
VPN
↓
Company Security
↓
Internet
```

Copy

All traffic can be inspected centrally.

Key clue



"All traffic must be inspected" → FULL TUNNEL



Hard Example 2 – DMVPN

A company has:

- 1 headquarters
- 300 branch offices
- Many branches need to communicate with each other
- Administrators don't want to manually configure hundreds of individual VPN tunnels.

Question

Which Cisco technology is appropriate?

Answer

☐ DMVPN

Why?

DMVPN is designed for **scalable dynamic VPN connectivity**.

Instead of manually configuring every tunnel:

```
HQ
//|\
//|\
Branches
```

 Copy

DMVPN allows dynamic tunnel establishment between spokes when needed.

Key clue



Many branches + dynamic tunnels = DMVPN





FINAL EXAM CHEAT SHEET

If you have very little time, memorize **these 15 things**:

1.

VPN = secure tunnel over an untrusted network.

2.

Remote access = user → company.

3.

Site-to-site = network → network.

4.

IPsec = secures IP traffic.

5.

IKEv2 = negotiates IPsec security.

6.

AnyConnect = Cisco remote-access VPN client.

7.

SSL VPN = remote access using TLS/SSL.

8.

Clientless VPN = browser-based access without installing a VPN client.

9.

DMVPN = scalable dynamic VPN for many branches.

10.

FlexVPN = flexible Cisco IKEv2 VPN framework.

11.

Split tunnel = only selected/company traffic uses VPN.

12.

Full tunnel = all traffic uses VPN.

13.

AES-256-GCM = modern strong encryption choice.

14.

PFS = protects past sessions if long-term keys are later compromised.

15.

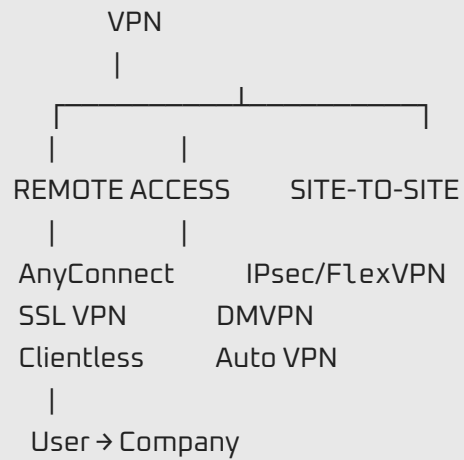
FTD = firewall + VPN + security/threat inspection.



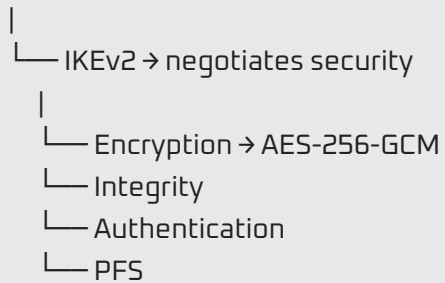


One-Minute Memory Map

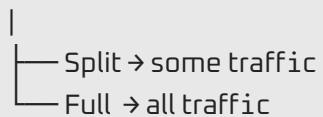
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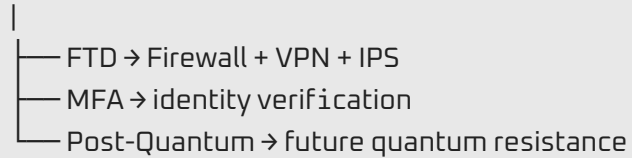
IPsec



TUNNELING



SECURITY



☆ Most likely exam questions

1. What is a VPN?
2. Difference between remote-access and site-to-site VPN?
3. What is IPsec?
4. What is the role of IKEv2?
5. What is AnyConnect?
6. What is DMVPN and why is it scalable?
7. Difference between split and full tunneling?
8. What is PFS?
9. What is clientless SSL VPN?
10. What is the purpose of AES-256-GCM?
11. What is FTD?
12. What is quantum-resistant cryptography?

Super-short memory trick:



□ IPsec protects → IKEv2 negotiates → AnyConnect connects users → DMVPN connects branches → FTD protects the edge → Full tunnel sends everything → Split tunnel sends selected traffic.